

EE579 Project Checkpoint 1: Project Idea and Literature Review

Jia Belanger, Brandon Rathburn, Kelly Wu

I. INTRODUCTION

A. Research question

Sloped, granular terrains pose challenges for wheeled locomotors. Wheel accelerations that require friction forces that are higher than what an angled, granular surface can provide, encounter slippage and struggle with rolling and effective locomotion.

Successful traversal of diverse terrain types, including granular slopes, is critical for data collection in planetary exploration. Rovers such as Opportunity, Spirit, and Curiosity each encountered significant mobility challenges on soft, granular surfaces, resulting in immobilization or subsequent rerouting [1]. Meanwhile, extensive testing is required to minimize the sinkage and slippage of lunar rover wheels on sloped terrains [2], highlighting the need for reliable and robust wheeled locomotion on sloped, granular media.

Thus, this paper aims to (1) identify a new wheeled gait pattern that combines whleg (wheel-leg) locomotion and whleg adduction/abduction angle oscillations and (2) assess the effectiveness of this gait in robot locomotion on sloped granular media.

B. Literature review

Previously, it has been demonstrated extensively that sinusoidal, snake-like gait patterns are highly effective on granular terrain. On sloped, granular terrains, particularly, side-winding, sinusoidal gaits have been shown to be a highly effective means of locomotion [3].

However, traveling sinusoidal and cyclic gaits depend on geometric shape changes over dynamics that are force-limited, meaning that the motion and structural integrity of the gait is limited by a maximal force. Thus, sinusoidal gaits are viable for low-power robots, but struggle with scalability in larger or heavier vehicles necessary for extraterrestrial exploration [4]. Wheeled, dynamic gaits allow for a heavier payload, but in turn struggle in granular terrains.

Furthermore, the introduction of a slope to granular surfaces adds complexity to previously examined findings of Resistive Force Theory (RFT) [5], which was developed to predict robots movement on flat, static, granular mediums. While this paper postulates that the key to understanding and improving inclined movement may be dependent on the angle of intrusion, the original RFT model possesses a limitation: it does not account for the shifting physics of an incline.

Research has been done on wheel-leg robots on flat surfaces and common indoor obstacles like stairs [6]. Although these

Mini-Whegs are able to travel at speeds of 3 BL/s and clear 15cm high stairs, other terrain experiments are limited. Even with robots that can move on granular media, there has been success with "half-circle" legs, most commonly associated with the RHex robot [7], and also with sand-swimming robots using a swimming gait [8]. However, there is not a lot of room for flexibility when it comes to the modularity of these half-circle legs themselves. Regarding sand-swimming robots, their leg designs are based off of biological animals, but do not challenge the idea of the use of whegs.

Prior works in improving wheeled locomotion on granular and sloped terrains include NASA's Resource Prospector 15 (RP15) MiniRover that introduces the "RS gait", which combines wheeled and legged crawling strategies to prevent and overcome granular media hindrances as well as traverse successfully over poppy seeds imitating granular media [9]. In the face of getting "stuck" in granular media, the RP15 would instead transition to using its wheel attached legs to "climb" out of the granular pit in a crawling gait. On the other hand, WhlegRunner, which tests 3-7 spoke wheels on various dry and damp sand environments [10], finds that sparse point contact whegs have an improved performance on all granular media tested in comparison to dense point contact whegs that more closely resemble wheels. However, studies limited to wheel centric locomotion designs on granular media have yet to find means of recovering from wheel entrapment or slippage without transitioning to an alternative legged locomotion modes.

Our proposed study addresses these knowledge gaps by exploring how active rear abduction and adduction can be integrated into a whleg-based system. We aim to combine the "swimming" logic of biological models with the whleg structure to discover if such orientation changes may overcome the frictional failures.

C. Hypothesis

We believe that adding a degree of freedom (DoF) to the wheel may increase mobility by allowing the gait to dynamically vary wheel movement that may help "wiggle" out of stuck positions and achieve an improved performance overall on granular media. Therefore, we hypothesize that a tri-whleg robot can utilize active abduction/adduction in its rear whegs during its forward gait to achieve a higher overall velocity on inclined granular media as compared to a normal whleg rotation gait.

REFERENCES

- [1] C. Li and K. Lewis, "The need for and feasibility of alternative ground robots to traverse sandy and rocky extraterrestrial terrain," *Advanced Intelligent Systems*, vol. 5, no. 3, p. 2100195, 2023. [Online]. Available: <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/aisy.202100195>
- [2] H. Nakashima, H. Fujii, A. Oida, M. Momozu, H. Kanamori, S. Aoki, T. Yokoyama, H. Shimizu, J. Miyasaka, and K. Ohdoi, "Discrete element method analysis of single wheel performance for a small lunar rover on sloped terrain," *Journal of Terramechanics*, vol. 47, no. 5, pp. 307–321, 2010. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0022489810000212>
- [3] H. Marvi, C. Gong, N. Gravish, H. Astley, M. Travers, R. L. Hatton, J. R. Mendelson, H. Choset, D. L. Hu, and D. I. Goldman, "Sidewinding with minimal slip: Snake and robot ascent of sandy slopes," *Science*, vol. 346, no. 6206, p. 224–229, Oct. 2014. [Online]. Available: <http://dx.doi.org/10.1126/science.1255718>
- [4] Q. Lan, Z. Wang, L. Ding, H. Yang, H. Gao, L. Richter, and Z. Deng, "Dem simulation and continuation algorithm of granular physical field for planetary wheel-terrain interaction," *Powder Technology*, vol. 433, p. 119197, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0032591023009804>
- [5] T. Zhang and D. I. Goldman, "The effectiveness of resistive force theory in granular locomotion," *Physics of Fluids*, vol. 26, no. 10, p. 101308, 10 2014. [Online]. Available: <https://doi.org/10.1063/1.4898629>
- [6] R. Schroer, M. Boggess, R. Bachmann, R. Quinn, and R. Ritzmann, "Comparing cockroach and whegs robot body motions," in *IEEE International Conference on Robotics and Automation, 2004. Proceedings. ICRA '04. 2004*, vol. 4, 2004, pp. 3288–3293 Vol.4.
- [7] E. Moore, D. Campbell, F. Grimmering, and M. Buehler, "Reliable stair climbing in the simple hexapod 'rhex'," in *Proceedings 2002 IEEE International Conference on Robotics and Automation (Cat. No.02CH37292)*, vol. 3, 2002, pp. 2222–2227 vol.3.
- [8] R. D. Maladen, Y. Ding, P. B. Umbanhowar, and D. I. Goldman, "Undulatory swimming in sand: experimental and simulation studies of a robotic sandfish," *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 793–805, 2011. [Online]. Available: <https://doi.org/10.1177/0278364911402406>
- [9] S. Shrivastava, A. Karsai, Y. O. Aydin, R. Pettinger, W. Bluethmann, R. O. Ambrose, and D. I. Goldman, "Material remodeling and unconventional gaits facilitate locomotion of a robophysical rover over granular terrain," *Science Robotics*, vol. 5, no. 42, p. eaba3499, 2020. [Online]. Available: <https://www.science.org/doi/abs/10.1126/scirobotics.aba3499>
- [10] H. Bagheri, V. Jayanetti, H. R. Burch, C. E. Brenner, B. R. Bethke, and H. Marvi, "Mechanics of bipedal and quadrupedal locomotion on dry and wet granular media," *Journal of Field Robotics*, vol. 40, no. 2, pp. 161–172, 2023. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22121>